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Time taken 20 mins

Grade 11 out of 20 (55%)

Question 1

Complete

Mark 1 out of 1

How do Docker image layers contribute to the containerization process?

Select one:

- ☐ A. They slow down Docker builds and increase disk usage
- ☐ B. Each layer originates from the next layer, reducing reusability
- ☐ C. Layers decrease reusability and increase disk usage
- ☒ D. Layers speed up Docker builds, increase reusability, and decrease disk use

The correct answer is: Layers speed up Docker builds, increase reusability, and decrease disk use

Question 2

Complete

Mark 0 out of 1

In the context of Docker, what is a swarm?

Select one:

- ☒ A. A collection of standalone containers
- ☐ B. A single Docker host in swarm mode
- ☐ C. Multiple Docker hosts running in swarm mode
- ☐ D. A specific type of Docker image

The correct answer is: Multiple Docker hosts running in swarm mode

Question 3

Complete

Mark 1 out of 1

What does a Docker image contain?

Select one:

- ☐ A. Only application code
- ☐ B. Libraries and tools, but not dependencies
- ☒ C. Application code, libraries, tools, and dependencies
- ☐ D. Only writable layers for user modifications

The correct answer is: Application code, libraries, tools, and dependencies

Question 4

Complete

Mark 1 out of 1

What does application virtualization allow?

Select one:

- ☐ A. Running multiple operating systems on a single server
- ☒ B. Running applications on different operating systems without modification
- ☐ C. Creating virtual desktop environments
- ☐ D. Simulating hardware functionality for testing purposes

The correct answer is: Running applications on different operating systems without modification

Question 5

Complete

Mark 0 out of 1

What does Docker do to maintain the desired state of a service in a swarm?

Select one:

- ☐ A. Halts the service when a worker node becomes unavailable
- ☐ B. Schedules tasks on other nodes when a worker node becomes unavailable
- ☒ C. Transfers tasks to standalone containers when a worker node becomes unavailable
- ☐ D. Promotes a worker node to a manager when tasks are not running

The correct answer is: Schedules tasks on other nodes when a worker node becomes unavailable

Question 6

Complete

Mark 0 out of 1

What does libcontainer use to create an isolated workspace for containers?

Select one:

- ☐ A. Dockerfiles
- ☐ B. Namespaces
- ☒ C. Docker Compose
- ☐ D. Container images

The correct answer is: Namespaces

Question 7

Complete

Mark 0 out of 1

What has Docker replaced LXC with for managing containers? **

Select one:

- ☒ A. LXD
- ☐ B. libvirt
- ☐ C. libcontainer
- ☐ D. libdocker

The correct answer is: libcontainer

Question 8

Complete

Mark 1 out of 1

What is a key advantage of containerization for developers?

Select one:

- ☐ A. Slower application deployment
- ☐ B. Limited portability across platforms
- ☒ C. Faster and more secure application creation and deployment
- ☐ D. Dependency-related bugs and errors

The correct answer is: Faster and more secure application creation and deployment

Question 9

Complete

Mark 1 out of 1

What is a key difference between standalone containers and swarm services?

Select one:

- ☐ A. Standalone containers can only run on swarm managers
- ☒ B. Only swarm managers can manage a swarm, while standalone containers can be started on any daemon
- ☐ C. Standalone containers cannot run on Docker hosts participating in the swarm
- ☐ D. Swarm services can only be managed by worker nodes

The correct answer is: Only swarm managers can manage a swarm, while standalone containers can be started on any daemon

Question 10

Complete

Mark 1 out of 1

What is containerization in the context of software development?

Select one:

- ☐ A. Creating a virtual machine (VM) with all dependencies
- ☒ B. Bundling software code with only the necessary OS libraries and dependencies
- ☐ C. Deploying applications exclusively on physical servers
- ☐ D. Developing code without any dependencies

The correct answer is: Bundling software code with only the necessary OS libraries and dependencies

Question 11

Complete

Mark 0 out of 1

What is the nature of Docker image layers?

Select one:

- ☒ A. Read-write files
- ☐ B. Writable layers for user modifications
- ☐ C. Read-only files
- ☐ D. Unchangeable files

The correct answer is: Read-only files

Question 12

Complete

Mark 1 out of 1

What is the primary purpose of a Docker image in the context of containerization?

Select one:

- ☐ A. Executing code directly on the host machine
- ☒ B. Acting as a set of instructions to build a Docker container
- ☐ C. Serving as a snapshot of a running container
- ☐ D. Creating a separate file for each Docker container

The correct answer is: Acting as a set of instructions to build a Docker container

Question 13

Complete

Mark 0 out of 1

What is the primary purpose of hardware virtualization?

Select one:

- ☐ A. Creating a virtual version of a computer network
- ☐ B. Abstracting physical storage resources
- ☐ C. Mimicking the functionality of physical computers with virtual machines
- ☒ D. Combining data from various sources into a single view

The correct answer is: Mimicking the functionality of physical computers with virtual machines

Question 14

Complete

Mark 0 out of 1

What is the role of Control Groups (cgroups) in libcontainer?

Select one:

- ☒ A. Managing container images
- ☐ B. Creating isolated workspaces
- ☐ C. Enforcing limits and constraints on resource usage
- ☐ D. Defining container environment variables

The correct answer is: Enforcing limits and constraints on resource usage

Question 15

Complete

Mark 1 out of 1

What is virtualization?

Select one:

- ☐ A. Creating a physical version of an operating system
- ☒ B. Creating a virtual version of various computing components
- ☐ C. Simulating hardware functionality to enhance performance
- ☐ D. Managing network resources without virtualization

The correct answer is: Creating a virtual version of various computing components

Question 16

Complete

Mark 1 out of 1

What roles can a Docker host perform in a swarm?

Select one:

- ☐ A. Only a manager
- ☐ B. Only a worker
- ☒ C. Both a manager and a worker
- ☐ D. Neither a manager nor a worker

The correct answer is: Both a manager and a worker

Question 17

Complete

Mark 1 out of 1

Which resources are typically isolated by libcontainer using namespaces?

Select one:

- ☐ A. Container names
- ☒ B. Process ID, network, IPC, filesystem
- ☐ C. Docker image layers
- ☐ D. Container environment variables

The correct answer is: Process ID, network, IPC, filesystem

Question 18

Complete

Mark 0 out of 1

Which resources can be limited by Control Groups (cgroups) in libcontainer?

Select one:

- ☒ A. Docker daemon resources
- ☐ B. Disk space, CPU, memory
- ☐ C. Container networking
- ☐ D. Container process IDs

The correct answer is: Disk space, CPU, memory

Question 19

Complete

Mark 1 out of 1

Which type of virtualization abstracts physical storage resources?

Select one:

- ☐ A. Network virtualization
- ☒ B. Storage virtualization
- ☐ C. Data virtualization
- ☐ D. Desktop virtualization

The correct answer is: Storage virtualization

Question 20

Complete

Mark 0 out of 1

What is a pod in Kubernetes and what does it do?

- ☒ a. All of the above
- ☐ b. One or more containers grouped together to share resources and run as a unit
- ☐ c. Collection of logic circuitry that evaluates a computer's operational code
- ☐ d. Collection of physical IT components that supports a group of containers

The correct answer is: One or more containers grouped together to share resources and run as a unit